

Probability & Statistics – Enhanced Study Notes

Improved PDF version with readable formulas, revision notes, and extra practice questions.

Original Notes

PROBABILITY & STATISTICS

Complete Study Notes

SPPU — Second Year Engineering | 2024 Pattern



5 Units | Formulas | Solved Examples | Quick Reference

UNIT I: Probability Fundamentals and Set Theory (5 Hours)

This unit covers the mathematical foundations of probability — from set algebra to conditional probability and Bayes' Theorem.

1.1 Algebra of Sets — Quick Definitions

Key Set Laws (Always Useful)

1.2 Definitions of Probability

Classical Definition

Used when all outcomes are equally likely (e.g., fair coin, fair die, well-shuffled deck).

Axiomatic Definition (Kolmogorov's Axioms)

These are the 3 rules that EVERY valid probability must satisfy:

From these axioms, we can derive:

- $P(\emptyset) = 0$
- $P(A') = 1 - P(A)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $0 \leq P(A) \leq 1$

Frequentist (Empirical) Definition

Used for real-world experiments when classical equally-likely assumption doesn't hold.

1.3 Addition Rule of Probability

1.4 Conditional Probability

'What is the probability of A, given that B has already occurred?'

This leads to the Multiplication Rule:

1.5 Independence of Events

Two events A and B are INDEPENDENT if knowing one doesn't change the probability of the other.

1.6 Bayes' Theorem ★★★

One of the most powerful results in probability. Used to update our beliefs when new evidence arrives.

Setup — Total Probability Theorem First

If $B_1, B_2, \dots, B_n$ are mutually exclusive and exhaustive events (they partition S):

Bayes' Theorem Formula

UNIT II: Random Variables and Expectation (7 Hours)

A random variable assigns a number to each outcome of a random experiment.

2.1 Types of Random Variables

2.2 Probability Mass Function (PMF) — Discrete RV

2.3 Probability Density Function (PDF) — Continuous RV

2.4 Expectation (Mean) and Variance

2.5 Higher-Order Moments

2.6 Moment Generating Function (MGF)

2.7 Chebyshev's Inequality and Markov's Inequality

UNIT III: Distributions and Functions of Random Variables (8 Hours)

This is the most formula-heavy unit. We cover all major probability distributions and their properties.

3.1 Summary of Key Distributions

3.2 Binomial Distribution $B(n, p)$

Models the number of successes in n INDEPENDENT trials, each with $P(\text{success}) = p$.

3.3 Poisson Distribution $P(\lambda)$

Models the number of events occurring in a fixed interval of time/space, when events happen at constant average rate λ and independently.

Poisson Approximation to Binomial

When n is large ($n \geq 50$) and p is small ($p \leq 0.1$), use Poisson with $\lambda = np$.

3.4 Geometric Distribution $G(p)$

3.5 Exponential Distribution $\text{Exp}(\lambda)$

Models time between events in a Poisson process. Memoryless like the geometric.

3.6 Normal Distribution $N(\mu, \sigma^2)$ ★★★

The most important distribution in statistics. Symmetric, bell-shaped. Many natural phenomena follow it.

Normal Approximation to Binomial

When n is large ($np \geq 5$ and $nq \geq 5$), Binomial can be approximated by Normal with continuity correction.

3.7 Gamma Distribution $\Gamma(\alpha, \beta)$

3.8 Bivariate Normal and Joint Distributions

UNIT IV: Sampling and Estimation (5 Hours)

Statistics moves from populations (theory) to samples (real data). We use sample data to estimate population parameters.

4.1 Key Vocabulary

4.2 Sampling Distributions

A sampling distribution is the distribution of a statistic (like $\bar{x}$) across many samples.

4.3 Central Limit Theorem (CLT) ★★★

4.4 Point Estimation

Properties of Good Estimators

4.5 Maximum Likelihood Estimation (MLE)

4.6 Method of Moments (MOM)

4.7 Confidence Intervals ★★★

A confidence interval (CI) gives a range where we believe the true parameter lies.

UNIT V: Hypothesis Testing and Decision Theory (5 Hours)

Hypothesis testing lets us make decisions about populations based on sample evidence.

5.1 Framework for Hypothesis Testing

5.2 Types of Errors

5.3 Neyman-Pearson Lemma

5.4 Z-Test (Large Sample / Known σ)

5.5 t-Test (Small Sample / Unknown σ)

5.6 Chi-Square Test (χ^2)

5.7 F-Test (Comparing Two Variances)

5.8 Two-Sample Tests

MASTER FORMULA SHEET — All Units

High-Yield Revision & Practice Questions

Unit I Extra Practice

- State and prove the addition rule of probability.
- Solve two Bayes' theorem problems involving medical testing.
- Differentiate independent and mutually exclusive events.

Unit II Extra Practice

- Find PMF, CDF, mean and variance for a given discrete RV.
- Derive $E[X]$ and $\text{Var}(X)$ from an MGF.
- Apply Chebyshev's inequality for different k values.

Unit III Extra Practice

- Compare Binomial, Poisson and Normal distributions.
- Solve one problem each on Binomial, Poisson and Exponential distributions.
- Use Z-tables for Normal distribution probabilities.

Unit IV Extra Practice

- Derive the Central Limit Theorem statement.
- Construct 90%, 95% and 99% confidence intervals.
- Find required sample size for given margin of error.

Unit V Extra Practice

- Perform Z-test, t-test and Chi-square test.
- Explain Type I and Type II errors with examples.
- Choose the correct statistical test for a given scenario.